Total No. of Printed Pages:04

SUBJECT CODE NO: - RNP-8006
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group A] Mech, Civil
Examination January/February 2025
Engineering Mechanics

[Time: 3:00 Hours] **[Max. Marks:60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Answer the following questions. **20**

1. In a concurrent force system, the forces:
   a) Do not have a common point of application
   b) Have a common point of application
   c) Act along parallel lines
   d) Are always equal in magnitude

2. A non-coplanar force system is....
   a) All forces lie in the same plane
   b) Forces are at different angles in space
   c) All forces act parallel to one another
   d) All forces have a common point of action

3. The unit of moment is?
   a) Nm b) mN
   c) m/s d) m/s^2

4. Which method is used to analyse forces in a truss?
   a) Direct integration method
   b) Method of Joint
   c) Moment distribution method
   d) Newton's method

5. In a plane truss, if a member is under compression, the force in the member is:
   a) Positive b) Negative
   c) Zero d) Undefined

6. A force system is said to be in equilibrium if:
   a) The sum of all external forces is zero
   b) The sum of all moments about any axis is zero
   c) Both a and b
   d) None of the above

7. The moment of inertia of a body is a measure of:
   a) Its resistance to rotation
   b) Its resistance to linear motion
   c) Its ability to do work
   d) Its energy

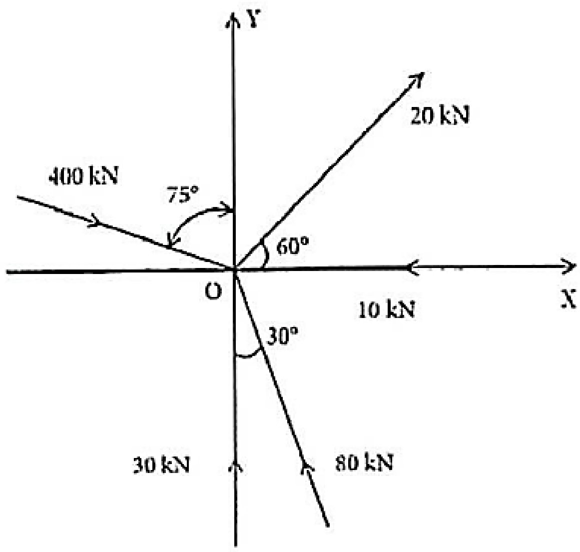
8. Which of the following is true about a system in equilibrium?
   a) The sum of all forces and the sum of all moments must be zero.
   b) Only the sum of all forces must be zero.
   c) Only the sum of all moments must be zero.
   d) None of the above.

9. the trajectory of a projectile is:
   a) A straight line. b) A parabola.
   c) A circle. d) An ellipse.

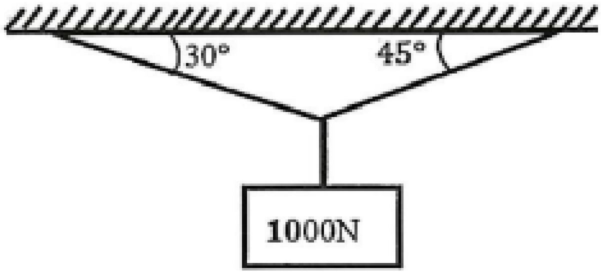
10. Which law of motion states that for every action, there is an equal and opposite reaction?
    a) Newton's First Law b) Newton's Second Law
    c) Newton's Third Law d) Law of Universal Gravitation

SECTION B

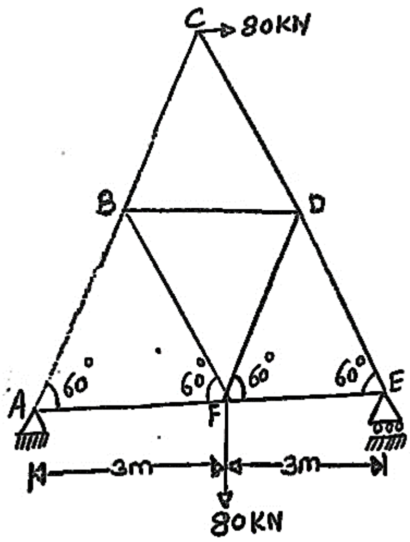
Q2 a) Find magnitude and direction of given concurrent force system. **05**



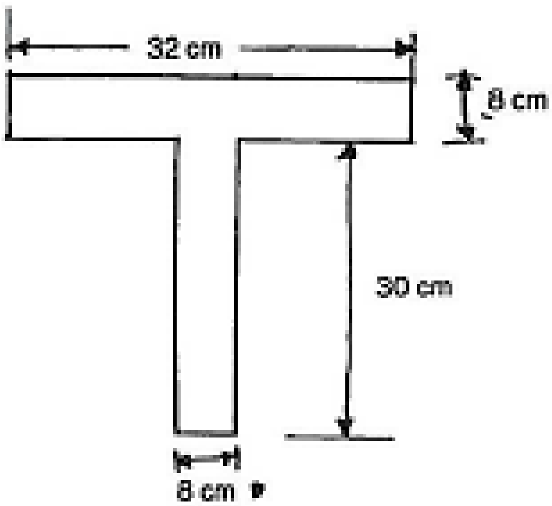
b) A weight of 1000N is attached by two strings calculate the tension in the string. **05**



Q3 Find out the magnitude of forces in given truss. **10**



Q4 Determine the centroid of T-section shown in figure below about X and Y axis. **10**



Q5 Displacement equation of a particle is given by, **10**

$$S= 40t + 16t^2 - 3t^3$$

Find:

a) Velocity and acceleration at start

b) Time when particle reaches its maximum velocity

c) The maximum velocity of the particle.

Q6 a) A ball is thrown with a velocity of 150 m/s at an angle of elevation of 60° with the horizontal. **05**

Find:

a) Maximum height reached by the ball

b) Time of flight

c) Range of ball

b) A bullet is fired from a gun with an initial velocity of 150 m/s to hit it a target. The target is located at a horizontal distance of 1800 m and 350 m above the gun. Determine minimum angle of projection so that the bullet will hit the target. **05**

Q7 An automobile of mass 1200 kg moves down a 25° incline at a speed of 40 kmph, when the brakes are applied causing a constant braking force of 7 kN. Determine the distance travelled by the Automobile as it comes to rest. **10**